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Unit 5 : Session 12 : SLO 1 | ARM Addressing and Instruction Encoding

I. Multiple Choice

1. B) LDR R0, =0x40 the literal-pool pseudo-instruction loads a constant.
2. B) Post-indexed the base register is updated after the memory access.
3. C) Opcode + operands + condition encoding covers all three fields.

II. Fill in the Blanks

1. before pre-indexed addressing updates the base register before access.
2. LDR literal constants are loaded with LDR Rd, =constant.
3. shift scaled register offset applies a shift (e.g. LSL) to the index register.

III. True or False

1. TRUE.
2. FALSE encoding identifies opcode, operands and condition fields, not just the opcode.
3. TRUE.

IV. Match the Following

- A) LDR 3) Load data from memory.
B) STR 1) Store register to memory.
C) Pre-indexed 4) Base updated before access.
D) Literal LDR 2) Load from constant pool.

V. Numerical Answers

1. 0x400C 0x4000 + 12.
2. ADD R4, R2, R1 #1 cond=1110 (AL), I=0, opcode=0100 (ADD), S=0, Rn=0010 (R2), Rd=0100 (R4), operand2=R1 #1 binary 1110 0000 1000 0010 0100 0000 0000 0001 = 0xE0824001.
3. 0xE2821005 #1 cond=1110, I=1 (immediate), opcode=0101 (ADC), S=0, Rn=R2, Rd=R1, operand2=#5 #1 ADC R1, R2, #5.